

Ruth Elvin Harianja

Data Scientist

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Work Experience

Data Scientist Intern
Feb 2026 - Apr 2026

BKPSDM Kota Medan

- Built a Python preprocessing pipeline to remove duplicate attendance records, standardize attendance statuses, and engineer temporal and geospatial features.
- Applied spatial and spatiotemporal DBSCAN clustering to identify attendance outliers and used the Haversine formula to calculate distances from estimated office locations.
- Implemented a rule-based scoring system combining geographic distance, clustering outliers, attendance notes, and location-status discrepancies to classify anomaly risk as low, medium, or high.
- Developed an interactive Streamlit dashboard with employee- and department-level analysis, geographic maps, attendance timelines, approval summaries, and CSV exports to support attendance review.

Quality Assurance Engineer
Oct 2022 - Aug 2024

PT. Saka Uniti Indonesia

- Work closely with the Product team, including PMs, Designers, Developers, and other QA engineers right from the inception of features.
- Prepare test case documents based on documented requirements.
- Run test cases and analyze the results.
- Report bugs and errors to the development team.
- Create test scripts for running automation testing using Selenium.

Software Quality Assurance
Oct 2021 - Sept 2022

PT. Bumi Amartha Teknologi Mandiri

- Execute test cases and report bugs and errors to the developer.
- Work vertically on each feature to test the user experience and user flows on the UI (mobile application).
- Regression test in the SIT environment and smoke test in the Beta application before releasing to production.

Education

Master's Degree in Data Science and Artificial Intelligence
2024 - 2026
University of North Sumatra

Bachelor of Applied Science in Software Engineering Technology
2017 – 2021
Del Institute of Technology

Personal Skills

- Programming Languages: Python, Java, SQL.
- Machine Learning & AI: Scikit-learn, TensorFlow/Keras, PyOD (Isolation Forest, LOF, ECOD), Random Forest, Logistic Regression, SHAP (XAI), Genetic Algorithm, Perceptron/ANN.
- Data Science & Analytics: Pandas, NumPy, Statsmodels, Exploratory Data Analysis (EDA), Feature Engineering, Time Series Forecasting (LSTM, ARIMA).
- Geospatial Analytics: Spatial DBSCAN, ST-DBSCAN, Haversine Distance.
- Web Development & Data Visualization: Flask, Streamlit, Folium, Plotly.
- Software Testing & Security: Cypress, Selenium, Postman, Burp Suite.
- Databases & Backend Platforms: MySQL, PostgreSQL, Firebase.
- Development & Project Tools: Git, Jira.
- Business Platforms: Odoo ERP, MoEngage.